

BIOINFORMATICS BOOTCAMP

ASSESSMENT 01

Bioinformatics is often described as an interdisciplinary field. Explain why Bioinformatics requires knowledge from multiple disciplines. Discuss the role of biology, computer science, mathematics, and statistics in solving biological problems.

Answer:

Bioinformatics deals with the storage of biological data, the retrieval of data, analysis, and the interpretation of data. It inculcates multiple fields such as Biology, Mathematics, Statistics, and Computer Science.

Knowledge in Biology helps one to be well-versed in biological data such as DNA, RNA, and other genomes.

Since databases are required in bioinformatics, computerized archives are necessary for biological data storage and data organization, so that it will be much easier for information to be retrieved through a variety of search criteria. This is the role of Computer Science.

This integrates Mathematics and Statistics since scientists and researchers can now use multiple tools and software, prediction methods, algorithms, and statistical methods to predict results. This will enable researchers to cut down costs and reduce time spent in wet laboratories.

Rather than spending years to sequence DNA or RNA, bioinformatics makes use of all these fields to reduce the time spent in the lab, the trial and error associated with experimentation, and the high cost involved. Hence, drugs and vaccines do not take a long time to be produced. With the evolution of diseases and other pandemics, a shorter time frame can be used to design and develop drugs and vaccines through the interdisciplinary actions of all these combined fields.

An example is the COVID-19 pandemic. Without all these fields coming together, a vaccine for the virus would still not have been developed because laboratory experimentation would still have been ongoing. Bioinformatic databases helped to sequence the biological data of the virus and search for identical sequences of the coronavirus gene, perform a comparative analysis, and computationally predict, to design and develop a vaccine.

Question 1

A researcher has discovered a DNA sequence that may be associated with a rare genetic disorder. Before conducting laboratory experiments, they want to determine whether this sequence has been reported previously. Which biological database(s) would you recommend

for this purpose? Justify your answer by explaining the type of information each database provides and how it would assist the researcher.

Answer:

1. GenBank:

Since it is one of the largest public databases of DNA sequences, it has the most complete collection of annotated nucleic acid sequence data for almost every organism, which includes genomic DNA.

GenBank receives sequences produced in laboratories throughout the world and allows other scientists to manually submit data and other research findings from their labs into the database. It notes down the date it was published, the accession number, the sequence length, the type of molecule, the source of the organism, information about the particular organism, and other vital information that this particular researcher might need to determine whether the DNA sequence of this rare genetic disorder has ever been reported.

Hence, this researcher will be able to search for identical sequences submitted by other researchers worldwide using BLAST. Through that, he will be able to compare the query DNA sequence with other sequences in the database. He will also be able to have access to links to published and cited works and related databases related to this DNA sequence.

At the end, this researcher will be able to determine whether the DNA sequence of this rare genetic disorder has already been deposited in a public database.

Question 2

A scientist is studying a disease-causing bacterium and wants to compare its genome with those of related bacterial species to identify conserved genes. Which biological database(s) would you recommend for this analysis? Explain why these databases are appropriate and what information they provide.

Answer:

1. GenBank:

This is the most appropriate because it contains one of the largest repositories of nucleotide sequences. It has a complete bacterial genome derived from thousands of species. It will allow the scientist to retrieve genome sequences, which he can use to compare the disease-causing bacterium with those of related species. It will also enable him to retrieve associated annotations for comparative analysis because GenBank provides information on gene annotations, protein-coding regions, taxonomy information, and a complete genome sequence. It also provides links to related

literature and other NCBI resources. The scientist will be able to compare nucleotide sequences of bacterial species and use that information to detect conserved genes.

Question 3

A biotechnology company has isolated a protein from a newly identified microorganism and wants to investigate whether similar proteins exist in other species. Which biological database(s) would you recommend, and what information would you expect to obtain from each database? Explain your reasoning.

Answer:

1. UniProt:

For protein sequences and functional annotations, UniProt is the best database. There are experimentally approved and computationally predicted proteins obtained from a variety of organisms, hence, this biotechnology company will be able to make use of UniProt to tell whether other species contain proteins similar to the one they have isolated in this newly identified organism. UniProt provides the sequences of proteins and their names, their biological roles, and references to other databases. With this information, the biotechnology company will be able to identify whether similar proteins exist in other species.

2. Protein Data Bank (PDB):

Data on proteins are available as both sequences and structures. PDB contains 3-dimensional protein structures that have been determined experimentally. This will enable the biotechnology company to identify whether structurally similar proteins exist, and it will also be able to understand the protein's potential functions.

Question 4

A scientist has identified an unknown protein sequence and wants to determine its function. Which biological database(s) would you recommend for this purpose? Justify your answer by explaining the type of information each database provides.

Answer:

1. Protein Data Bank (PDB):

PDB contains 3-dimensional protein structures that have been determined experimentally. To determine the function of the unknown protein sequence, PDB is appropriate because the structure of a protein is related to its function. PDB also provides information on binding sites and active

sites, which can help to figure out the function of this unknown protein, in case sequence similarity is low, before he decides to conduct experiments.

2. UniProt:

UniProt is a protein sequence and functional annotations database. It contains both experimentally approved information about proteins and computationally predicted information about proteins. It provides information on protein sequences, their names and functions, and enzyme information. Hence, the scientist will be able to identify proteins that have similar sequences to that of this unknown protein sequence, get to know their known functions, and use that information to predict the functions of this unknown protein.

Question 5

You have been provided with an unknown biological sequence but are not told whether it is DNA, RNA, or protein. Describe the step-by-step approach you would follow to identify the sequence using biological databases. Explain which databases you would use at each stage and why.

Answer:

1. I will first take note of the composition of the sequence. DNA contains Adenine, Thymine, Cytosine, and Guanine nucleotides. RNA contains Adenine, Uracil (in place of Thymine), Cytosine, and Guanine nucleotides. Protein sequences have amino acid letters like A, N, D, E, F, and others.
2. After I have been able to determine which biological sequence it is, I will use the various databases and search tools.
3. I will search nucleotide databases like GenBank, if the sequence is a DNA or RNA sequence, to compare this unknown biological sequence that has been provided to me with the millions of known sequences. This will help me know the particular organism and species the sequence is from, identify the specific gene, identify whether this particular sequence has ever been published, and find other previously published sequence records.
4. I will also search protein databases like PDB and UniProt, if the sequence is a protein. First of all, UniProt will help me identify the name of the protein, functions, cellular location, and enzyme activity. Afterwards, PDB will help me to identify the 3-dimensional structure of the protein, and hence, I will be able to identify its active site, binding site, and other structural similarities it has with other proteins.